

IEOR 4404 – Fall 2025

Practice Midterm (Structural Variant A)

Problem 1 (Discrete ITM + A/R on a conditioned Poisson). Let X be a Poisson random variable conditioned to be *even*:

$$\mathbb{P}(X = 2i) = \frac{e^{-\lambda} \lambda^{2i} / (2i)!}{\sum_{j=0}^{\infty} e^{-\lambda} \lambda^{2j} / (2j)!}, \quad i = 0, 1, 2, \dots, \quad \lambda > 0.$$

- (a) Derive an inverse transform method (ITM) to generate X *on the even support* by exploiting the recursion $\mathbb{P}(X = 2(i+1)) = \frac{\lambda^2}{(2i+1)(2i+2)} \mathbb{P}(X = 2i)$. Give an $O(\mathbb{E}[X])$ algorithm that accumulates the even masses until exceeding a uniform U .
- (b) Show that the following procedure is an acceptance–rejection (A/R) algorithm with a *constant* acceptance test: sample $Y \sim \text{Pois}(\lambda)$ and accept if Y is even, setting $X = Y$; otherwise repeat. State the dominating pmf g , the target pmf f , the constant C , and the acceptance probability.

Problem 2 (Continuous ITM + tail check for A/R). Consider the *log-logistic* (a.k.a. Fisk) distribution on $(0, \infty)$ with cdf

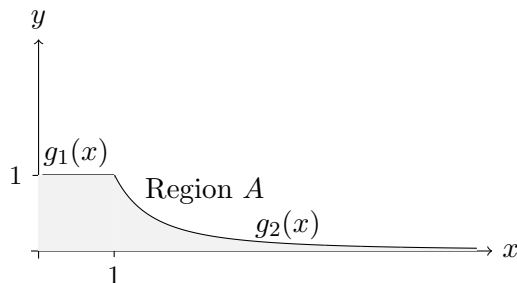
$$F(x) = \frac{x^\beta}{1 + x^\beta}, \quad \beta > 0.$$

- (a) Derive an ITM sampler for X by inverting F .
- (b) You insist on using A/R with a proposal from the *same family* but parameter $\tilde{\beta} > 0$. Determine for which direction of the inequality between β and $\tilde{\beta}$ the ratio $f(x)/g(x)$ is bounded on $(0, \infty)$. Conclude precisely when the A/R scheme is valid and when it fails.

Problem 3 (Uniform points in a piecewise region via ITM). Let A be the region under g_1 on $[0, 1]$ and under g_2 on $[1, \infty)$:

$$g_1(x) = 1 \cdot \mathbf{1}\{0 < x \leq 1\}, \quad g_2(x) = x^{-2} \cdot \mathbf{1}\{1 \leq x < \infty\}.$$

- (a) Compute the total area of A and the marginal cdf of X for a point (X, Y) dropped uniformly in A .
- (b) Give an ITM sampler for X , then conditionally generate $Y \mid X$ uniformly on $[0, g(X)]$.



Problem 4 (Composition method with a different kernel). Consider the nonnegative random variable X with density

$$f_X(x) = \int_0^\infty \frac{2t}{(1+xt)^3} e^{-t} dt, \quad x \geq 0.$$

Derive a simulation algorithm by interpreting the integrand as a mixture: first generate $Z \sim \text{Exp}(1)$, then generate $X \mid Z = z$ by inverting the conditional cdf corresponding to the kernel $\frac{2z}{(1+xz)^3}$.